

Chapter 35 - The HOST Command

HOST is the BASIC verb that lets a program ask the Intuition Engine appliance to perform a system-level action: bring up network configuration, update the installed system packages, reboot the machine, or power it off. Five subverbs are recognised.

HOST is not a tokenizer keyword. It is matched as raw ASCII at statement dispatch in `exec_line`, so the four letters `H O S T` are stored in the program text exactly as typed. In expression context and in explicit `LET` assignments the same letters are still an ordinary identifier and may be used as a variable name; the statement form is recognised only when `HOST` begins a statement.

35.1 Subverbs

Form	What it does
HOST NET	Open the network configuration screen.
HOST UPDATE	Update the installed system packages.
HOST REBOOT	Reboot the appliance.
HOST POWEROFF	Power off the appliance.
HOST HELP	Print the list of <code>HOST</code> subverbs to the terminal.

HOST on its own (no subverb) is the same as `HOST HELP`. A following colon (`HOST:`) is also the help form, so the keyword can appear at the start of a colon-separated multi-statement line without arguments.

Any other word after `HOST` produces a syntax error and the statement aborts before the appliance is touched.

35.2 What each subverb does

35.2.1 HOST NET

Opens the appliance's network configuration screen. The user picks a network, enters credentials, and confirms. The screen exits when the user dismisses it. BASIC resumes the next statement when the screen closes.

35.2.2 HOST UPDATE

Updates the installed system packages on the appliance. The action runs through an authenticated channel; the appliance asks the user to confirm before the change is applied. If the user cancels, the statement returns without error and execution continues with the next BASIC statement.

35.2.3 HOST REBOOT

Reboots the appliance. The running program stops, the machine restarts, and the BASIC prompt comes back when the boot sequence finishes. A program that calls `HOST REBOOT` never sees the next statement on the same boot.

35.2.4 HOST POWEROFF

Powers the appliance off. As with `REBOOT`, the running program does not return; the next time BASIC runs is after a manual power-on.

35.2.5 HOST HELP

Prints a fixed list of the subverbs (NET, UPDATE, REBOOT, POWEROFF, HELP) with a one-line description of each. The text comes from a string in the BASIC ROM and is not configurable.

35.3 The register block

HOST is built on a four-register block at 0xF1400. Every register is 32-bit. For the command, trigger, and status registers only the low byte carries information; the exit register is a full 32-bit unsigned value, and reading any of its four byte lanes returns the appropriate byte of that value.

Offset	Name	R/W	Width	Meaning
+0x00	command	W	byte	Subverb enum. 1 = NET, 2 = UPDATE, 3 = REBOOT, 4 = POWEROFF, 0 = none.
+0x04	trigger	W	byte	Writing a non-zero value fires the command currently in the command register.
+0x08	status	R	byte	Current state of the command. See below.
+0x0C	exit	R	32-bit	Exit code from the underlying action, valid once status reaches a terminal state. Read as a full 32-bit word; the four lane addresses +0x0C-+0x0F return the four bytes of the same value.

The status register reads:

Value	State	Terminal?
0	running	no
1	OK	yes
2	error	yes
3	cancelled by user	yes
4	disabled (appliance refuses the command)	yes
5	idle (no command has been fired)	yes

A command starts in running when the trigger is written and moves to one of the terminal values when the action finishes. The exit register is only meaningful once the status is terminal.

35.4 The BASIC protocol

The BASIC verb implements this sequence:

1. Write the subverb enum to +0x00.
2. Write 1 to +0x04 to fire it.
3. Poll the byte at +0x08. While it reads 0, keep polling.
4. Once the byte is non-zero, check it against 1 and 3. 1 (OK) and 3 (cancelled by user) are success-shaped; both return control to BASIC with no error.
5. Any other terminal value (2, 4, or an unexpected 5) raises ?FC ERROR (illegal function call).

The disabled state (4) appears when the appliance has the system-action bridge turned off entirely. From BASIC this looks the same as any other failure: ?FC ERROR, then the prompt.

35.5 Calling the block directly

The block is a normal MMIO region. The IE64, IE32, M68K, and x86 CPUs reach it at 0xF1400 directly and drive it the same way the BASIC verb does: write the enum, write the trigger, poll status, read exit.

The 6502 and Z80 do **not** have a path to this block. Their CPU adapters translate the small \$F000-\$FFFF aperture onto the 0xF0000-0xF0FFF page only; addresses in the 0xF1xxx page have no 16-bit alias on either adapter, so the HOST appliance block is unreachable from bare 8-bit code. Programs running on the 6502 or Z80 ask a larger CPU to drive HOST for them through the coprocessor channel (Chapter 31), or run the verb from BASIC.

From IE64:

```
la      r2, 0xF1400
li      r1, #2                ; UPDATE
store.b r1, (r2)
li      r3, #1
store.b r3, 4(r2)

.poll:
load.b  r4, 8(r2)
li      r5, #0
beq     r4, r5, .poll        ; running
; R4 now holds the terminal status
load.l  r6, 12(r2)          ; R6 = full 32-bit exit code
```

The poll is a tight loop in the example above. Real programs should run it from a timer-driven interrupt or yield between reads so that the rest of the system continues to schedule.

35.6 What HOST does not do

HOST is the only BASIC verb that affects the appliance as a whole. It does not:

- expose a generic shell;
- run arbitrary commands;
- read or write files on the volume (use LOAD / SAVE and the File I/O block, Chapter 34);
- send packets onto the network (the network is configured by HOST NET, but no BASIC verb sends raw network traffic);
- return any text output to the program. The action prints status to the appliance's own surface; the BASIC program only sees the terminal status code described above.

HOST is the on-ramp for system administration from inside a program. Anything else the program needs from the outside world goes through the file-I/O block, the terminal, or one of the input or media chapters.